

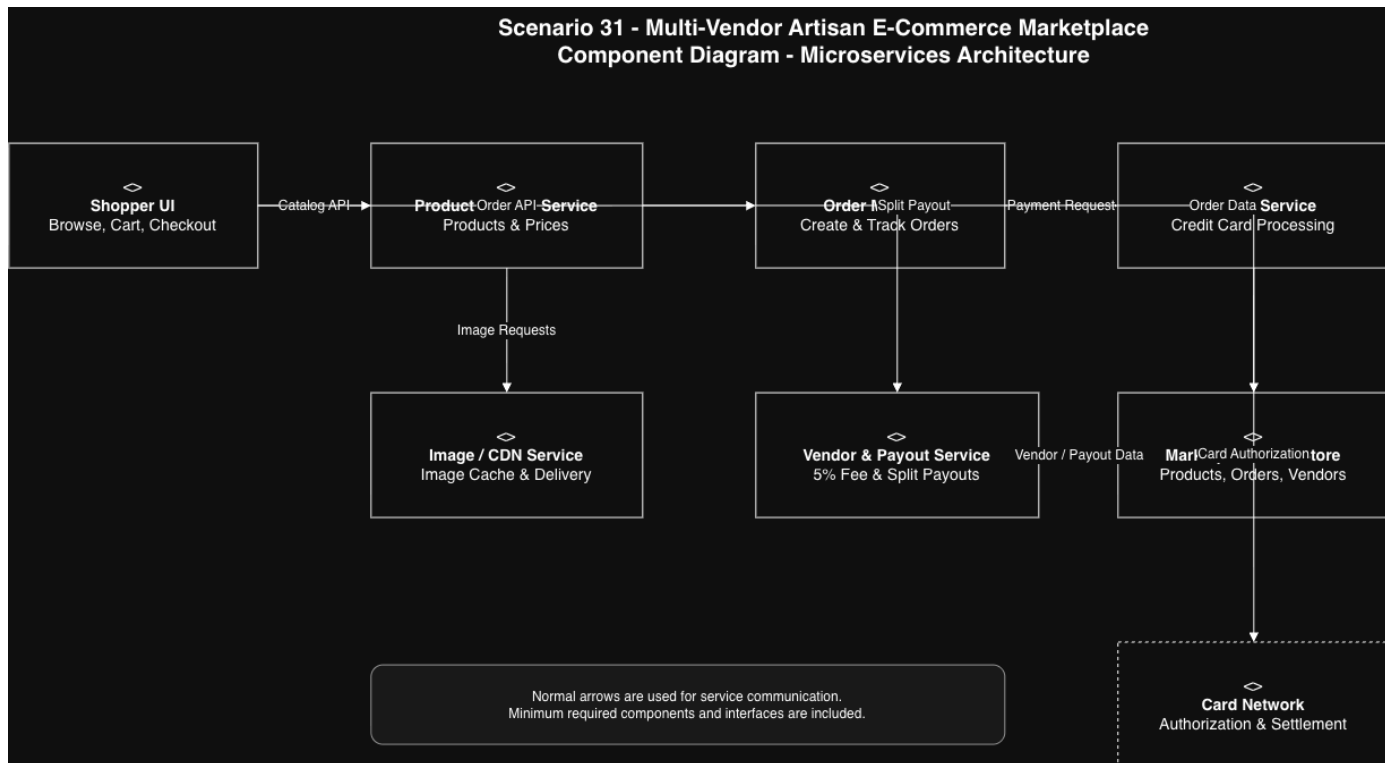
Lab 3 - Architecture Selection & Written Justification

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Scenario #31: Multi-Vendor Artisan E-Commerce Marketplace

1. UML Component Diagram



2. Architecture Selection- Multi-Vendor Artisan E-Commerce Marketplace

Architecture Selection

We chose Microservices Architecture for the Multi-Vendor Artisan E-Commerce Marketplace.

Why this architecture

1. Independent vendor and commerce functions: The marketplace naturally separates into services such as Product Catalog, Order Manager, Payment Service, and Vendor & Payout Service. Each service can be changed or deployed independently, reducing coupling as the marketplace grows across many artisans and products.
2. Scalability and fault isolation: Product browsing and image delivery can receive much higher traffic than payout processing. Microservices allow the heavily used catalog/CDN path to scale independently while isolating failures so a problem in one service does not require the entire marketplace to be taken down.

Security Advantage

Payment processing is isolated in a dedicated Payment Service that communicates with the external card network through a defined interface. This separation limits the scope of sensitive payment logic, supports stricter access controls around payment operations, and keeps card-processing concerns away from general catalog and storefront components.

Performance Benefit

The catalog and high-resolution product-image path can be optimized independently, including CDN caching for frequently requested images. This directly supports the scenario requirement for image delivery with a target load time below 500 ms, while the payment and payout services can scale separately during checkout peaks.

Key Trade-off

Microservices introduce operational complexity, network calls, and data-consistency challenges. For this scenario, the benefits of independent scaling, service isolation, and clear separation of catalog, ordering, payment, and vendor payout responsibilities outweigh those costs.